LINEAR CORRELATION AND REGRESSION BETWEEN THE METEOROLOGICAL DATA AND THE ELECTRICITY DEMAND OF THE DUBROVNIK REGION IN A SHORT-TERM SCALE

by

Anamarija FALKONI [a*] ***and Goran KRAJAČIĆ*** [b]

[a] Maritime Department, University of Dubrovnik, Dubrovnik, Croatia
[b] Department of Energy, Power Engineering and Environment, Faculty of Mechanical Engineering and Naval Architecture, University of Zagreb, Zagreb, Croatia

Review paper
DOI: 10.2298/TSCI151209157F

Intermitted patterns of solar and wind generation cause insecurity in the power system supply since they depend on weather conditions. The aim of this work is to perform linear regression and correlation between solar radiation, wind speed, air temperature and electricity demand of the Dubrovnik region. Gained results could help in system energy planning with a high share of renewable energy sources in the electricity production. All of the data are collected for consecutive three year period, years 2012, 2013 and 2014, in the 10 minute time step. Results of the correlation of each of the parameters individually in between the years show slightly variations between the distributions, providing representative linear relation in between the years for all the data, except the wind speed data. Correlations are also done between all of the parameters for each year separately, based on the mean monthly values. Result showed good relation with negative correlation between solar radiation and wind speed, as well as good relation with positive correlation between solar radiation and electricity demand. The same correlations are done in the 10 minute time step and including time system delay. The results indicate significant decrease in correlation coefficient value and it is less possible that they can be pronounced with linear regression line. Calculations of the correlation and regression, based on the 10 minute time step for summer and winter period separately, gained slightly better results in relation between parameters than the ones including the whole year data.

Key words: *wind speed, solar radiation, air temperature, electricity demand, correlation coefficient*

Introduction

European Union (EU) has issued a series of laws and policies which seek to encourage a higher energy production from renewable energy sources (RES) in order to reduce pollution, enhance the exploitation of its own natural resources and to achieve the independence of the energy sector. European countries accepted targets of the Directive 2009/28/EC [1], so called the Renewable Energy Directive, which makes the EU one of the most advanced groups of nations in promoting renewable energy production. According to the EU, the Republic of Croatia has published its own Energy Strategy, which among other goals seeks to achieve 20% of RES, including large hydropower plants, in the final energy consumption by 2020. Croatia has set up a goal to maintain the level of 35% of a share of electricity genera-

* Corresponding author; e-mail: anamarija.sare@unidu.hr

tion from renewable energy sources, including large hydropower plants, in overall electricity consumption until 2020. Energy sectors in Croatia participate with approximately 75% in total greenhouse gas emissions. Croatia is currently importing about 50% of its energy demand, where oil and oil products are represented with 50% and natural gas with 25%. Projections show a global increase in the energy consumption for about 50% until 2030, which indicates the need for new sources of energy in the future. In order to increase security and supply competitiveness, Croatia is decided for elastic energy system with various renewable energy sources (wind farms, biomass power plants, small hydro power plants, solar power plants, waste to energy power plants, geothermal power plants) and energy supply courses and improvement of energy efficiency. The goal is to use 84 PJ of renewable energy sources in 2020 [2]. Negative environmental impact caused by the energy production from the conventional energy sources, exhaustible nature of fossil fuels and rise of oil prices emphasize the need to integrate RES into the power systems, which have promising economic, social and environmental benefits.

Different methods in achieving stable integration of solar and wind electricity production in the energy system

Higher interest in renewable production in recent years has led to an increase in the installed photovoltaic (PV) systems and wind power plants, causing insecurity in the system power supply. Solar and wind energy production have insecure and random generation patterns as they depend on weather and climate changes. Because of the stochastic behaviour of wind speed and solar radiation, wind and solar energy production are hard to predict. Previous studies have shown that the stochastic behaviour of wind speed and solar radiation can be described based on the analyses of meteorological data of wind speed, sunshine duration and solar radiation. It has been estimated that the Weibull and Gamma distributions can be appropriate to represent wind speed while the Beta distribution is approved to be suitable for the relative sunshine duration [3]. The analyses based on the European meteorology data derived the same results [4]. Intermittent RES electricity production can cause an excess or a lack of production, resulting in the overloads of the system power grid. The results from the previous studies showed that the production system has better performance when solar and wind energy productions are combined instead of alone PV or wind generation system. It is concluded that the optimal mix and appropriate placement and sizing of RES could lead to many positive effects in the system, like improving voltage stability, reducing lower network losses and enhancing loadability of lines [5].

Many studies have been done for the analyses in this field, providing different solutions to maintain security of the power system and to achieve its flexibility and stability. Results in study [6] showed that the high variable RES integration in the power system causes high uncertainty. The authors suggested that the RES electricity production should be combined with storage facilities, combined heat and power systems and power plants, which fuel can be easily stored, as well as the cross-border transmission capacities, which can contribute to the efficient utilization of RES. In study [7], the model of microgrid is presented, which aims to match electricity production from RES and the electricity demand by combining storage facilities and electricity prices. Hybrid energy systems, including PV and wind energy systems with battery storage, are shown to be more reliable and cost effective, due to the complementary nature of these two resources. Another study [8] provides an extensive literature overview of 140 journal articles of stand-alone and grid connected hybrid systems. In [9] two hybrid systems were analysed. First one included PV panels, biomass gasifier and storage

in a battery bank and in the other one the existing configuration is added with wind generator. First system showed better results and seemed to be able to cover the residential electricity demand profile in a flexible and a reliable way with a very low level of losses. Concept of a model for a future building of self-sustainable house, with solar power production and electric vehicles (EV), in region of Croatia, was presented in [10]. Results of three different scenarios were presented. The third scenario with EV charging control and home demand side management can consist of 100% renewable energy production.

Study [11] is presenting a model of forecast errors of wind power generation with battery storage. Correlation analysis between electricity prices and wind generation was done in order to achieve flexibility and stability of the German power system. Wind and solar energy generation forecast and the power system operation mainly focuses on the immediate short-term of seconds to minutes, a short-term of hours up to two days and medium-term of two to seven days. Short terms are usually statistically based. Authors in [12] presented a detailed overview of current methods and recent advances in wind power forecasting. Several forecasting models, for wind speed and power on a different time scale, were discussed in [13-15]. Another study in [16] is providing a classification according to forecasting data; wind speed forecasting (indirect method) and wind power forecasting (direct method). The authors suggest combining different physical and statistical models to achieve good results both in long term and short term forecasting, as well as further research on the practical application of the methods, not only in theoretical. Results of a new model for wind power forecasting, according to the measured data of wind speed and power obtained from a wind farm in northern Sicily, are presented in [17]. One new method is proposed in [18] for short-term wind power and speed forecasting. Study [19] gave the results of the solar forecasting model in short and medium-term. Other studies, [20] and [21], provided the results for incorporating wind speed and solar radiation forecasting models. Besides solar and wind forecasting, study [22] presented a model for electrical load forecasting which had achieved a great success in industry recently. Author concluded that the hourly temperature is critical for the hourly load forecasting and that their system needs to improve the performance with this type of data. Loads in weekends show different patterns and are more difficult to predict and further work needs to be done to improve the accuracy in weekend load prediction.

Wind and solar electricity production profiles should complement each other as much as possible in order to provide stability of the power system with high electricity production from RES and minimise the need for additional storage and production units. Large-scale complementarity between the monthly data of wind and solar electricity production showed good correlation between two sources of energy, while medium-scale of daily data provides medium-term complementarity and a small-scale of one hour data shows even less complementarity [23]. Besides the complementarity between wind and solar electricity production, analyses should be done in complementarity between RES production and electricity demand profiles. Analysis of the complementarity between the wind production and the electricity demand profile done in [24], shows weak correlation between both profiles and ramping issues. Correlation analyses between electricity load and weather conditions were obtained for three buildings providing the results of response of the office building electricity consumption to the weather conditions. Results showed that a temperature and a specific humidity have significant influencing factors for electricity consumption, while the wind speed statistical testing results were not significant [25]. Study [26] provided results of correlation analyses between the electricity load curve and statistical data for wind speed and solar irradiation for a typical year and a typical day. The results provided a good correlation for a typi-

cal year and a typical day between RES potential and electricity demand, which encourages higher RES integration into the power system and electricity production in more environmentally friendly manner than fossil fuel power plants. Results also showed complementarity between wind and solar energy and hydroelectricity, which is the region's main energy resource. Stability of the power system, with a high production from RES, depends on the geographical system size. Larger energy systems have fewer requirements for flexibility than the smaller ones [27]. Flexibility requirements of the European power systems will depend on the optimal mix of RES and their share, balancing area size, as well as aiming at transnational solutions as the most efficient way for large-scale integration of RES. Analyses of the operational timescales beyond 12 hours are relevant for developing a long-term storage system as a part of stability and flexibility for integrated RES. Flexibility requirements in the time horizon of minutes are important for the design of automatic generation control schemes. One way of achieving flexibility in the system will be presented within this work, where the calculations will be based on the 10 minute time data.

All of the previous presented studies gave some kind of solution in order to enhance the integration of RES into the power systems. Stability and security of those systems could be achieved by taking into consideration weather analyses and forecasting, storage facilities and electricity market. Good operation and management of the power systems with a high production from RES will be hard to achieve without using information and communication technologies (ICT-tools), which would help in control of the system energy flows. The entire electricity infrastructure and associated socio-technical systems including the system operation, supply, generation, consumers, transmission and distribution networks and market mechanisms will need to evolve to realize the cooperation and maintain security of the grid, so called smart grids [28-30]. Studies [31] and [32] presented standards for power utility communication which help to ensure a smooth and a reliable activity in the network. Beside the smart grids, power systems should be arranged as smart energy systems which use new available technologies and infrastructure that create new forms of flexibility and use RES in a more efficient way. The smart energy system includes all sectors of the energy system, providing flexibility by combining electricity, heat and transport sector, to compensate lack of flexibility due to the production of energy from RES [33, 34].

Aims of the research work

The Dubrovnik region has a high potential in RES, mainly high wind and solar potential. The aim for the future energy plan of the region's power system is to integrate high share of RES in the electricity production. Literature overview, provided within this work, presented different methods for stable integration of solar and wind power into the power system. Stability and flexibility of the system can be achieved through electricity storages, using forecast models, correlation analysis or arranging power system as a smart energy system. Linear correlation and regression will be used within this work to provide the results of complemetarity between solar and wind potential and electricity demand. Those results could be helpful in future energy system planning of the Dubrovnik region, as well as other regions. Literature provided within this work, dealing with the correlation analyses between the data on potential of RES and electricity demand, was based on the hourly and monthly time step. This work will provide the results of the linear correlation and regression analyses for the data on the 10 minute time step. Analyses based on the short-term scale could be meaningful for the future energy systems with an open electricity market based on the short-term scale, as well. The results from this study will be compared with the results from the previous studies.

Calculations are done between the selected parameters of solar radiation, wind speed, air temperature and electricity demand of the Dubrovnik region in a short-term scale, the 10 minute time horizon, and a medium-term scale, mean monthly data. Data for the selected parameters were collected for the years 2012, 2013 and 2014. Correlations are done for each of the parameters individually between the selected years in order to obtain results for the linear correlation and regression of the parameter distributions. The aim of this work is to provide the results of the linear correlation and regression for four different sets of data:

– correlation and regression of the 10 minute data for each of the parameter individually in between the years,
– correlation and regression in between the parameters for the consecutive three year period base on the mean monthly values,
– correlation and regression in between the parameters for the consecutive three year period base on the
  (1) 10 minute time step,
  (2) 10 minute time step including system time delay, and
– correlation and regression of 10 minute data in between the parameters for the consecutive three year period base on the
  (1) Winter period,
  (2) Summer period.

Methods

Previous study [35] on the energy plan of the Dubrovnik region showed that the integration of high share of RES and EV batteries in the electricity system still results in the critical excess of electricity production. These results indicate a demand for a larger capacity of storage facilities or new solutions to gain stability and flexibility of the system, in order to reduce a critical excess of electricity production. Gained results indicate the need for further future work in order to encourage the integration of RES into the power system and maintain its stability and flexibility. One of the solutions will be obtained within this work through the correlation and regression analyses in between RES potentials and electricity demand. Calculations of the linear correlation and regression will provide the results, based on which it can be estimated a complementarity between solar and wind potential and their complementarity with the electricity demand of the Dubrovnik region. The results could help in the future energy planning of the region's power system and enhance the integration of RES in the electricity production. Further work can be done, based on these results, in order to estimate optimal mix of RES in electricity production according to the electricity demand of the region. Studies [23-26], mentioned earlier in the text, have shown the results of the correlation analyses between the RES potential and electricity production and the electricity demand load on the monthly and the hourly bases. This work will go a step further and will provide the results of the correlation and regression analyses in the 10 minutes time horizon for different sets of selected data. Results of the calculations based on the 10 minute time step data can play a significant role for the possible future open energy market based on the 10 minute time step. Implementation of open energy market could also enhance the integration of RES into the power system, as well as help to reduce critical excess of electricity production.

Measuring and preparing the input data

The data of the parameters used in the analyses, including solar radiation, wind speed, air temperature and electricity demand, are collected for the years 2012, 2013 and 2014

and the calculations are done for the time step of t = 10 min. Data of solar radiation, wind speed and air temperature of the Dubrovnik region are provided by Croatian Meteorological and Hydrological Service for the years 2012, 2013 and 2014 in the 10 minute time horizon [36]. Measured data are gained from the meteorological station Dubrovnik. The device for measuring wind speed was anemometer with a performance of hemisphere and accuracy of ±5% (5-75 m/s), with a range of 0.5-75 m/s. The equipment for measuring solar radiation was CMP21 Pyranometer with the specifications provided in reference [37], CVF4 Ventilation Unit for global radiation measurements with the specifications provided in [38] and CM121 Shadow Ring for diffuse radiation measurements with the specifications provided in [39]. Air temperature was measured with Campbell-Stokes heliograph and Lambrecht thermograph. Heliograph measures radiations higher than 0.838 J/cm^2 per minute and when Sun is three degrees above the horizon. Thermograph measurement uncertainty is around 0.3 degree in the standard 10 degrees Celsius. Electricity demand data of the Dubrovnik region are provided by Elektrojug Dubrovnik – HEP ODS Ltd., the distribution system operator, for the years 2012, 2013 and 2014 in the 15 minute time horizon [40]. Electricity demand data were arranged in the 10 minute time horizon using linear interpolation.

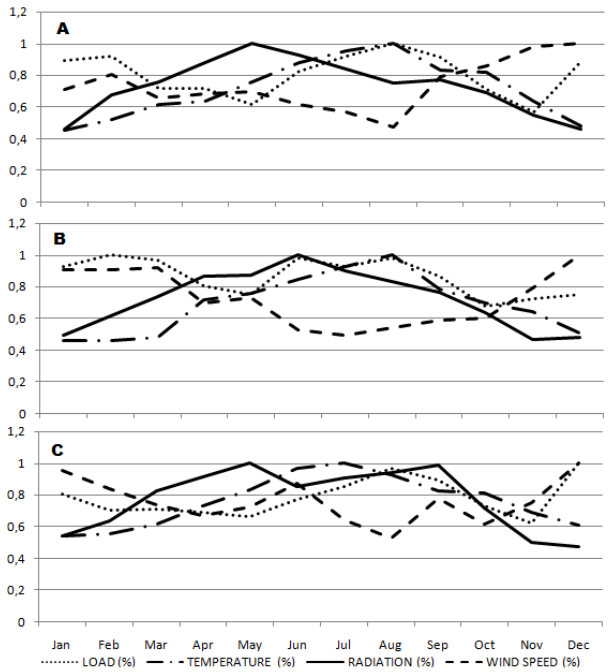

Figure 1. Mean monthly values normalized to their maximum value

Diagrams in fig. 1 show the distributions of mean monthly values of the selected parameters for each year normalized to their maximum value. A stands for distributions in 2012, B for 2013 and C for the year 2014. It can be concluded from the diagrams that the electricity demand depends on air temperatures. During the summer period, June, July, August and September, electricity demand increases as the temperature increases due to the need of cooling demand. During the winter period, January, February, November and December, electricity demand increases as the temperature decreases due to the need of heating demand.

Wind electricity production could be used to supply most of the electricity demand during the winter period due to the higher values of wind speed. On the other hand, solar electricity production could be used to supply most of the electricity demand during the summer period due to the high solar radiation. The problem with collected data of solar radiation, wind speed and temperature was that some of the data are missing. The data are missing mostly from the summer period of July and August. Solar radiation, for example, should reach its maximum during July and August, but because of the lack of data we have maximum during May and June. Results in tab. 1 show maximum and minimum mean monthly values for each of the parameters for every year.

Table 1. Values of the correlation coefficient between the selected variables with data based on the mean monthly values

	Load [kW]	Temperature [°C]	Radiation [Jcm^{-2}]	Wind speed [ms^{-1}]
Max (2012)	48,272.19	27.7	202.03	4.93
Min (2012)	28,460.92	7.43	41.61	2.46
Max (2013)	39,573.39	26.39	201.46	5.40
Min (2013)	24,636.00	9.74	41.75	2.58
Max (2014)	36,268.27	25.33	176.77	5.35
Min (2014)	23,537.71	11.58	31.72	2.69

Linear correlation and regression

The linear correlation and regression analyses are done for the different selected sets of data in order to establish the relationship between the parameters. First analysis (1) is done for each of the parameters individually to examine a linear relationship between the 10 minute data, t = 10 min, in between the years. Second analysis (2) is done in order to establish a linear relationship in between the parameters for each of the selected years separately based on the mean monthly values. Third set of data (3) were data of all the parameters arranged in 10 minute time step through the year, with the additional analysis which included system time delay between the electricity production and the demand. Electricity demand and air temperature are arranged for the period of t = 10 min and solar radiation and wind speed are taken for the time step of $t_1 = t + 4$ h. Since the meteorological data are different for different parts of the year, for example difference between the data of winter and summer period, the last analyses (4) are done for summer and winter periods. Summer period for each year was determined according to the highest temperature and highest peak electricity demand for three months in a row. Winter period for each year was determined according to the lowest temperature and highest peak electricity demand for three months in a row. The linear correlation and regression analyses are done for the data arranged in 10 minute time step between each of the parameters for two periods of each year separately.

Analyses are based on the model of simple linear regression and correlation providing the results of correlation coefficient, coefficient of determination and linear regression line between selected parameters. Calculations are done in STATISTICA, a comprehensive analytic, research and business intelligence tool used in business, data mining, science and engineering applications [41]. Calculations in STATISTICA, for this study, are based on basic formulas of linear correlation. Flow chart in fig. 2 is presenting a method used in this work. Formula in fig. 2 is presenting a basic formula of linear regression line gained from the calculations of the linear correlation were we have:

- $\widehat{y_i}$ – dependent variable,
- $\widehat{x_i}$ – independent variable,
- a – constant, the expected value of the dependent variable when the independent variable is zero, and
- b – regression coefficient shows the average change in the dependent variable caused by the change of the independent variable.

Dependent variable in other case can also be observed as an independent variable and a and b values will be different for that regression line. The coefficient of determination, r^2, is a key output of the regression analysis. It is interpreted as the proportion of the variance in the dependent variable that is predictable from the independent variable. The Pearson prod-

uct-moment correlation coefficient, r, illustrates a quantitative measure of some type of correlation and dependence, meaning statistical relationship between two or more random variables or observed data values.

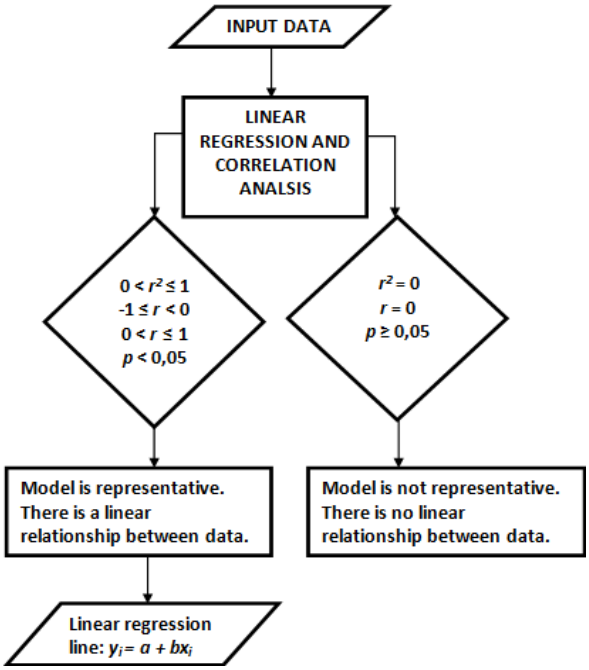

Figure 2. Flow chart

Results and discussion

Linear correlation and regression are done for the different sets of data based on the mean monthly values and 10 minute time step as it is explained in sub-section 1.2. The results of the calculations are provided in the following sub-sections.

Calculation results based on the first data set in 10 minutes time step for consecutive three year period

Calculations for the first data set (1) are split in four parts analysing the relationship between each of the parameters individually in between the years 2012, 2013 and 2014. Calculations are done for solar radiation, wind speed, air temperature and electricity demand data based on the 10 minute time step through the whole year. Results are obtained in tab. 2 with given values of r, r^2, a and b for each of the selected parameters in between the years, from which can be determined each of the linear regression line according to the equation given in fig. 2.

Variables x and y in the equation of linear regression line can be replaced with electricity demand (E_n, [kW]), solar radiation, (I_n, [Jcm^{-2}]), air temperature, (T_n, [°C]), and wind speed, (V_n, [ms^{-1}]), where n stands for the data of the years 2012, 2013 or 2014 and will have the same meaning in other calculations. Value N represents a number of data used in calculations. Results show the value of correlation coefficient r between two selected variables x and y and coefficient of determination r^2. Constant a and regression coefficient b in the equation of linear regression line are given for the case if y is the dependent variable or x is dependent variable. Equation of the linear regression line between variables x and y can be pronounced using values of constant a and regression coefficient b. Results of the calculations between the variables show significant relationship, $p < 0.05$, meaning there is a good probability that the observed relation between variables in the sample is a reliable indicator of the relation between the respective variables in the population. Value of correlation coefficient indicates significant linear correlation between the variables E_n, I_n and T_n, which means that their distributions in 10 minute time step slightly vary between consecutive three years period and they can be forecasted using linear regression line. Correlation coefficient for V_n distribution is close to 0 which means that the wind speed is hard to predict and data vary in between the years.

Table 2. Calculation results of the first data set in 10 minutes time step for consecutive three year period

x, y	Mean	Std. dv.	$r(x, y)$	r^2	p	N	$a; y$	$b; y$	$a; x$	$b; x$
Electricity demand										
E_{2012}	38,219.50	10,661.31								
E_{2013}	33,931.98	8,930.11	0.851455	0.724976	0.00	52,560	6,674,044	0,713195	3,726,995	1.016519
E_{2012}	38,219.50	10,661.31								
E_{2014}	28,474.17	7,059.98	0.838916	0.703779	0.00	52,560	7,241,920	0,555535	2,146,973	1.266851
E_{2013}	33,931.98	8,930.11								
E_{2014}	28,474.17	7,059.98	0.791839	0.627009	0.00	52,560	7,232,307	0,626013	5,412,514	1.001591
Solar radiation										
I_{2012}	106.5377	162.3647								
I_{2013}	103.1322	161.0447	0.828616	0.686605	0.00	34,822	15.57103	0,821880	20.38026	0.835408
I_{2012}	106.5377	162.3647								
I_{2014}	97.8802	158.2237	0.817296	0.667972	0.00	34,822	13.02809	0,796451	24.44705	0.838685
I_{2013}	103.1322	161.0447								
I_{2014}	97.8802	158.2237	0.768492	0.590580	0.00	34,822	20.01220	0,755030	26.57099	0.782194
Air temperature										
T_{2012}	17.16265	7.483789								
T_{2013}	17.19362	6.473829	0.872629	0.761481	0.00	48,667	4.23814	0,754865	−0.18167	1.008764
T_{2012}	17.16265	7.483789								
T_{2014}	17.14066	5.622084	0.810729	0.657282	0.00	48,667	6.68778	0,609048	−1.33547	1.079195
T_{2013}	17.19362	6.473829								
T_{2014}	17.14066	5.622084	0.794331	0.630962	0.00	48,667	5.28010	0,689823	1.51554	0.914672
Wind speed										
V_{2012}	3.552737	3.137801								
V_{2013}	3.774814	3.369657	0.103533	0.010719	0.00	47,248	3.379809	0,111183	3.188810	0.096409
V_{2012}	3.552737	3.137801								
V_{2014}	3.746781	3.135703	0.149070	0.022222	0.00	47,248	3.217529	0,148970	2.993831	0.149170
V_{2013}	3.774814	3.369657								
V_{2014}	3.746781	3.135703	0.072837	0.005305	0.00	47,248	3.490923	0,067780	3.481547	0.078272

Calculation results of the second set of data based on the mean monthly values

Analyses of the second set of parameters (2) are done for each year separately to establish linear relationship in between electricity demand, solar radiation, wind speed and air temperature data for each year separately. Data are based on the mean monthly values for each of the parameter and the results are shown in tab. 3. Calculations based on the mean monthly values are done in order to compare the results with results from previous studies.

Results in tab. 3, that are not marked bold, show that there is no linear relationship between the variables since their $p > 0.05$. Correlation results of E_{2014} *vs.* T_{2014} and E_{2014} *vs.* I_{2014} indicate good relationship between variables and are marked bold. Previous study done for Brazil [26] showed that $r = 0.46$ in case of E *vs.* I, while in case of E *vs.* V was $r = 0.29$. Good correlation results are provided for T_n *vs.* I_n, T_n *vs.* V_n and V_n *vs.* I_n. Study done for Italy [22] compared V *vs.* I on monthly bases and provided the results of correlation coefficient reaching values lower than −0.8 in several areas and for the nation-wide gained values were between −0.65 and −0.6, while the value of r was even closer to 0 for the daily based data. Linear regression line can be written using the equation of linear regression and the gained results.

Table 3. Calculation results of the second data set based on the mean monthly values for consecutive three year period

x, y	Mean	Std. dv.	$r(x, y)$	r^2	p	N	$a; y$	$b; y$	$a; x$	$b; x$
2012										
E_{2012}	38,221.26	6,703,026								
T_{2012}	17.67	7,157	0.278244	0.077420	0.381196	12	6.32	0.00	33,616.32	260.61
E_{2012}	38,221.26	6,703,026								
I_{2012}	113.13	57,820	0.323019	0.104341	0.305783	12	6.64	0.00	33,984.69	37.45
E_{2012}	38,221.26	6,703,026								
V_{2012}	3.59	0.980	−0.264313	0.069862	0.406446	12	5.06	−0.00	44,702.76	−1,807.95
T_{2012}	17.67	7,157								
I_{2012}	113.13	57,820	0.794745	0.631620	0.002009	12	−0.32	6.42	6.54	0.10
T_{2012}	17.67	7,157								
V_{2012}	3.59	0.980	−0.834147	0.695801	0.000743	12	5.60	−0.11	39.51	−6.09
I_{2012}	113.13	57,820								
V_{2012}	3.59	0.980	−0.812641	0.660386	0.001315	12	5.14	−0.01	285.03	−47.95
2013										
E_{2013}	33,945.51	5,087,649								
T_{2013}	17.57	6,024	0.203949	0.041595	0.524908	12	9.37	0.00	30,919.74	172.26
E_{2013}	33,945.51	5,087,649								
I_{2013}	114.51	61,956	0.459679	0.211304	0.132711	12	−75.51	0.01	29,622.99	37.75
E_{2013}	33,945.51	5,087,649								
V_{2013}	3.80	1,068	−0.226821	0.051448	0.478368	12	5.42	−0.00	38,051.79	−1,080.36
T_{2013}	17.57	6,024								
I_{2013}	114.51	61,956	0.903236	0.815835	0.000057	12	−48.67	9.29	7.51	0.09
T_{2013}	17.57	6,024								
V_{2013}	3.80	1,068	−0.862404	0.743740	0.000307	12	6.49	−0.15	36.05	−4.86
I_{2013}	114.51	61,956								
V_{2013}	3.80	1,068	−0.841384	0.707928	0.000602	12	5.46	−0.01	300.00	−48.80
2014										
E_{2014}	28,452.90	3,654,105								
T_{2014}	17.52	5,100	0.774233	0.599436	0.003117	12	−13.23	0.00	18,736.92	554.70
E_{2014}	28,452.90	3,654,105								
I_{2014}	104.55	55,722	0.637465	0.406361	0.025759	12	−172.04	0.01	24,082.46	41.80
E_{2014}	28,452.90	3,654,105								
V_{2014}	3.74	0.987	−0.477260	0.227777	0.116651	12	7.41	−0.00	35,068.73	−1,767.76
T_{2014}	17.52	5,100								
I_{2014}	104.55	55,722	0.845374	0.714657	0.000534	12	−57.23	9.24	9.43	0.08
T_{2014}	17.52	5,100								
V_{2014}	3.74	0.987	−0.806156	0.649888	0.001541	12	6.47	−0.16	33.11	−4.17
I_{2014}	104.55	55,722								
V_{2014}	3.74	0.987	−0.863273	0.745241	0.000298	12	5.34	−0.02	287,03	−48,76

Calculation results of the third set of data based on 10 minutes time step

The analyses of the third set of data (3) are done for each year separately to establish a linear correlation and regression in between the selected variables based on the 10 minute time horizon. The results are provided in tab. 4. Case A represents the results of the calcula-

tions done for the time step, t = 10 min, and case B represents the results for including system time delay, $t_1 = t + 4$ h, for each year separately.

Table 4. Calculation results of the third set of data based on 10 minutes time step

x, y	Mean	Std. dv.	$r(x, y)$	r^2	p	N	$a; y$	$b; y$	$a; x$	$b; x$
A – 2012										
E_{2012}	36,698.17	10,309.57								
T_{2012}	17.55	6.83	0.207789	0.043176	0.000000	39,188	12.50	0.000	31,194.94	313,546
E_{2012}	36,698.17	10,309.57								
I_{2012}	114,88	170.03	0.238412	0.056840	0.000000	39,188	−29.42	0.004	35,037.46	14,455
E_{2012}	36,698.17	10,309.57								
V_{2012}	3.52	3.10	−0.033600	0.001129	0.000000	39,188	3.89	−0.000	37,091.70	−111,908
T_{2012}	17.55	6.83								
I_{2012}	114.88	170.03	0.457128	0.208966	0.000000	39,188	−84.79	11.377	15.44	0.018
T_{2012}	17.55	6.83								
V_{2012}	3.52	3.10	−0.230647	0.053198	0.000000	39,188	5.35	−0.104	19.34	−0.509
I_{2012}	114.88	170.03								
V_{2012}	3.52	3.10	−0.060655	0.003679	0.000000	39,188	3.64	−0.001	126.60	−3.332
B – 2012										
E_{2012}	36,672.83	10,303.81								
T_{2012}	17.56	6.83	0.228995	0.052439	0.000000	39,164	11.99	0.0002	30,609.70	345.3601
E_{2012}	36,672.83	10,303.81								
I_{2012}	114.96	170.06	0.317798	0.100995	0.000000	39,164	−77.40	0.0052	34,459.38	19.2549
E_{2012}	36,672.83	10,303.81								
V_{2012}	3.52	3.10	−0.028810	0.000830	0.000000	39,164	3.84	−0.0000	37,010.14	−95.8866
T_{2012}	17.56	6.83								
I_{2012}	114.96	170.06	0.456913	0.208770	0.000000	39,164	−84.72	11.3734	15.45	0.0184
T_{2012}	17.56	6.83								
V_{2012}	3.52	3.10	−0.231189	0.053448	0.000000	39,164	5.36	−0.1048	19.35	−0.5102
I_{2012}	114.96	170.06								
V_{2012}	3.52	3.10	−0.060954	0.003715	0.000000	39,164	3.65	−0.0011	126.73	−3.3483
A – 2013										
E_{2013}	34,548.85	8,804,062								
T_{2013}	17.09	6,683	0.154528	0.023879	0.000000	44,189	13.03	0.0001	31,070.40	203.5715
E_{2013}	34,548.85	8,804,062								
I_{2013}	112.82	170,496	0.263732	0.069554	0.000000	44,189	−63.63	0.0051	33,012.43	13.6186
E_{2013}	34,548.85	8,804,062								
V_{2013}	3.80	3,418	−0.018643	0.000348	0.000089	44,189	4.05	−0.0000	34,731.43	−48.0282
T_{2013}	17.09	6,683								
I_{2013}	112.82	170,496	0.482973	0.233262	0.000000	44,189	−97.72	12.3215	14.95	0.0189
T_{2013}	17.09	6,683								
V_{2013}	3.80	3,418	−0.180600	0.032616	0.000000	44,189	5.38	−0.0924	18.43	−0.3532
I_{2013}	112.82	170,496								
V_{2013}	3.80	3,418	−0.116038	0.013465	0.000000	44,189	4.06	−0.0023	134.83	−5.7890
B – 2013										
E_{2013}	34,525.37	8,807,018								
T_{2013}	17.09	6,679	0.154885	0.023989	0.000000	44,159	13.04	0.0001	31,034.40	204.2413
E_{2013}	34,525.37	8,807,018								
I_{2013}	112.71	170,391	0.348371	0.121362	0.000000	44,159	−119.99	0.0067	32,495.86	18.0063
E_{2013}	34,525.37	8,807,018								
V_{2013}	3.80	3,418	−0.011112	0.000123	0.019543	44,159	3.95	−0.0000	34,634.21	−28.6312

Table 4. Continuation

x, y	Mean	Std. dv.	$r(x, y)$	r^2	p	N	$a; y$	$b; y$	$a; x$	$b; x$
T_{2013}	17.09	6,679								
I_{2013}	112.71	170,391	0.416649	0.173597	0.000000	44,159	−68.98	10.6297	15.25	0.0163
T_{2013}	17.09	6,679								
V_{2013}	3.80	3,418	−0.194796	0.037945	0.000000	44,159	5.51	−0.0997	18.54	−0.3806
I_{2013}	112.71	170,391								
V_{2013}	3.80	3,418	−0.115879	0.013428	0.000000	44,159	4.06	−0.0023	134.67	−5.7768
A – 2014										
E_{2014}	28,172.62	7,015,753								
T_{2014}	16.92	5,562	0.358931	0.128832	0.000000	47,376	8.90	0.0003	20,511.62	452.7414
E_{2014}	28,172.62	7,015,753								
I_{2014}	98.72	158,155	0.305787	0.093506	0.000000	47,376	−95.48	0.0069	26,833.53	13.5647
E_{2014}	28,172.62	7,015,753								
V_{2014}	3.83	3,205	−0.007734	0.000060	0.092306	47,376	3.93	−0.0000	28,237.39	−16.9304
T_{2014}	16.92	5,562								
I_{2014}	98.72	158,155	0.450734	0.203162	0.000000	47,376	−118.15	12.8165	15.36	0.0159
T_{2014}	16.92	5,562								
V_{2014}	3.83	3,205	−0.161061	0.025941	0.000000	47,376	5.40	−0.0928	17.99	−0.2795
I_{2014}	98.72	158,155								
V_{2014}	3.83	3,205	−0.072050	0.005191	0.000000	47,376	3.97	−0.0015	112.32	−3.5556
B – 2014										
E_{2014}	28,164.29	7,017,463								
T_{2014}	16.92	5,564	0.358288	0.128370	0.000000	47,237	8.92	0.0003	20,520.41	451.8583
E_{2014}	28,164.29	7,017,463								
I_{2014}	98.93	158,384	0.360612	0.130041	0.000000	47,237	−130.30	0.0081	26,583.62	15.9775
E_{2014}	28,164.29	7,017,463								
V_{2014}	3.82	3,200	0.003552	0.000013	0.440188	47,237	3.77	0.0000	28,134.55	7.7884
T_{2014}	16.92	5,564								
I_{2014}	98.93	158,384	0.386896	0.149689	0.000000	47,237	−87.37	11.0127	15.57	0.0136
T_{2014}	16.92	5,564								
V_{2014}	3.82	3,200	−0.178578	0.031890	0.000000	47,237	5.56	−0.1027	18.10	−0.3105
I_{2014}	98.93	158,384								
V_{2014}	3.82	3,200	−0.071292	0.005083	0.000000	47,237	3.96	−0.0014	112.40	−3.5286

Results of the correlation analyses between the parameters for each year indicate a significant decrease in value of the correlation coefficient in compare to the correlation results of the mean monthly values. The results in the case B for each year are slightly better than the ones in A. All of the results are marked as significant, except the one for E_{2014} *vs.* V_{2014}, but according to their low r values they are more likely to be unrepresentative to be express by a linear regression line. Linear regression line can be pronounced using coefficient a and b gained in tab. 4 and the equation of linear regression.

Calculation results of the fourth set of data based on 10 minutes time step for winter and summer period

Additional analysis has been done based on the data set of 10 minute time horizon (4), where the calculations are done to establish the relation between selected variables in two periods of each year. Diagrams in fig. 1 show that the wind and solar potential vary during the year. Solar radiation has its maximum during the summer period while wind speed has its maximum during the winter period. Accordingly, two periods of each year are selected to be analysed. Summer period for each year was determined according to the maximum air tem-

perature and maximum peak in electricity demand for three months in a row, June, July and August as show in fig. 1. Winter period is taken to be a period of three months in a row with a lowest air temperature and highest peak in electricity demand. For 2012 and 2014 it is taken to be for December, January and February, while for 2013 is taken to be for January, February and March as show in fig. 1.

Results of the linear correlation and regression are given in tab. 5, 6 and 7. Each table represents each year with case A and B. Case A shows the results for the selected summer period and case B shows the results for winter period.

Table 5. Correlation and regression results for summer and winter period for 2012

x, y	Mean	Std. dv.	$r(x, y)$	r^2	p	N	$a; y$	$b; y$	$a; x$	$b; x$
A – 2012										
E_{2012}	44,088.56	10,275.24								
T_{2012}	26.13	3.31	0.662773	0.439268	0.000000	9,074	16.73	0.000	−9,743.73	2,060,366
E_{2012}	44,088.56	10,275.24								
I_{2012}	193.18	213.12	0.318626	0.101522	0.000000	9,074	−98.20	0.007	41,121.05	15,362
E_{2012}	44,088.56	10,275.24								
V_{2012}	2.45	2.28	−0.029784	0.000887	0.004548	9,074	2.74	−0.000	44,417.65	−134,216
T_{2012}	26.13	3.31								
I_{2012}	193.18	213.12	0.537326	0.288720	0.000000	9,074	−712.05	34.647	24.52	0.008
T_{2012}	26.13	3.31								
V_{2012}	2.45	2.28	−0.009193	0.000085	0.381225	9,074	2.62	−0.006	26.16	−0.013
I_{2012}	193.18	213.12								
V_{2012}	2.45	2.28	0.167744	0.028138	0.000000	9,074	2.11	0.002	154.73	15,679
B – 2012										
E_{2012}	42,309.77	9,986,495								
T_{2012}	8.83	3,633	−0.287529	0.082673	0.000000	7,531	13.26	−0.000	49,289.32	−790,333
E_{2012}	42,309.77	9,986,495								
I_{2012}	53.47	94,390	0.036433	0.001327	0.001566	7,531	38.90	0.000	42,103.66	3,855
E_{2012}	42,309.77	9,986,495								
V_{2012}	4.48	3,084	0.060190	0.003623	0.000000	7,531	3.69	0.000	41,436.46	194,895
T_{2012}	8.83	3,633								
I_{2012}	53.47	94,390	0.434655	0.188925	0.000000	7,531	−46.25	11,292	7.94	0.017
T_{2012}	8.83	3,633								
V_{2012}	4.48	3,084	−0.139021	0.019327	0.000000	7,531	5.52	−0.118	9.56	−0.164
I_{2012}	53.47	94,390								
V_{2012}	4.48	3,084	−0.110865	0.012291	0.000000	7,531	4.67	−0.004	68.68	−3,393

Table 6. Correlation and regression results for summer and winter period for 2013

x, y	Mean	Std. dv.	$r(x, y)$	r^2	p	N	$a; y$	$b; y$	$a; x$	$b; x$
A – 2013										
E_{2013}	39,240.88	8,640,876								
T_{2013}	25.06	3,702	0.503655	0.253669	0.000000	10,610	16.60	0.000	9,772.74	1,175.673
E_{2013}	39,240.88	8,640,876								
I_{2013}	196.09	218,618	0.360576	0.130015	0.000000	10,610	−161,89	0.009	36,446.21	14,252
E_{2013}	39,240.88	8,640,876								
V_{2013}	2.66	2,301	−0.140073	0.019620	0.000000	10,610	4.13	−0.000	40,642.09	−526,119
T_{2013}	25.06	3,702								
										↓

Table 6. Continuation

x, y	Mean	Std. dv.	$r(x, y)$	r^2	p	N	a; y	b; y	a; x	b; x
I_{2013}	196.09	218,618	0.447643	0.200385	0.000000	10,610	−466.55	26.437	23.58	0.008
T_{2013}	25.06	3,702								
V_{2013}	2.66	2,301	0.036142	0.001306	0.000196	10,610	2.10	0.022	24.91	0.058
I_{2013}	196.09	218,618								
V_{2013}	2.66	2,301	0.000447	0.000000	0.963243	10,610	2.66	0.000	195.98	0.043
B – 2013										
E_{2013}	37,472.21	8,459,776								
T_{2013}	10.47	3,188	−0.112984	0.012765	0.000000	12,920	12.06	−0.000	40,610.95	−299,812
E_{2013}	37,472.21	8,459,776								
I_{2013}	60.90	110,088	0.030708	0.000943	0.000481	12,920	45.93	0.000	37,328.50	2,360
E_{2013}	37,472.21	8,459,776								
V_{2013}	5.05	4,090	0.034421	0.001185	0.000091	12,920	4.42	0.000	37,113.00	71,192
T_{2013}	10.47	3,188								
I_{2013}	60.90	110,088	0.408545	0.166909	0.000000	12,920	−86.79	14.108	9.75	0.012
T_{2013}	10.47	3,188								
V_{2013}	5.05	4,090	0.232899	0.054242	0.000000	12,920	1.92	0.299	9.55	0.182
I_{2013}	60.90	110,088								
V_{2013}	5.05	4,090	−0.139772	0.019536	0.000000	12,920	5.36	−0.005	79.88	−3,762

Table 7. Correlation and regression results for summer and winter period for 2014

x, y	Mean	Std. dv.	$r(x, y)$	r^2	p	N	a; y	b; y	a; x	b; x
A – 2014										
E_{2014}	34,259.62	7,440,386								
T_{2014}	24,78	2,711	0.531300	0.282280	0.000000	9,200	18.15	0.000	−1,867.68	1,458,034
E_{2014}	34,259.62	7,440,386								
I_{2014}	173.95	207,594	0.343010	0.117656	0.000000	9,200	−153.93	0.010	32,121.16	12,294
E_{2014}	34,259.62	7,440,386								
V_{2014}	2.89	2,108	0.045017	0.002026	0.000016	9,200	2.45	0.000	33,800.84	158,917
T_{2014}	24.78	2,711								
I_{2014}	173.95	207,594	0.552724	0.305504	0.000000	9,200	−874.69	42,321	23.52	0.007
T_{2014}	24.78	2,711								
V_{2014}	2.89	2,108	0.183272	0.033588	0.000000	9,200	−0.64	0.142	24.10	0.236
I_{2014}	173.95	207,594								
V_{2014}	2.89	2,108	0.144022	0.020742	0.000000	9,200	2.63	0.001	132.99	14,186
B – 2014										
E_{2014}	27,495.46	6,493,145								
T_{2014}	11.91	3,437	−0.339692	0.115391	0.000000	12,936	16.86	−0.000	35,138.66	−641,666
E_{2014}	27,495.46	6,493,145								
I_{2014}	39.49	77,695	0.105663	0.011165	0.000000	12,936	4.73	0.001	27,146.71	8,830
E_{2014}	27,495.46	6,493,145								
V_{2014}	5.18	4,046	0.113016	0.012773	0.000000	12,936	3.24	0.000	26,556.15	181,372
T_{2014}	11.91	3,437								
I_{2014}	39.49	77,695	0.224492	0.050397	0.000000	12,936	−20.95	5,074	11.52	0.010
T_{2014}	11.91	3,437								
V_{2014}	5.18	4,046	0.036559	0.001337	0.000032	12,936	4.67	0.043	11.75	0.031
I_{2014}	39.49	77,695								
V_{2014}	5.18	4,046	−0.064810	0.004200	0.000000	12,936	5.31	−0.003	45.94	−1,245

The results are different when comparing two periods for each year. It can be seen from the results that the solar radiation has better correlation with electricity demand during summer period. Wind has a positive correlation with electricity demand, which means when wind increases demand increases, as well indicating that the wind could be a good energy resource during the winter time. Although r value is close to 0 and the relation between variables cannot be pronounced with linear regression line. Wind has a good positive correlation during the summer period in 2014 indicating that wind could be a good energy resource during the summer time, as well.

Conclusions

The aim of this work was to provide the results of the correlation and regression analyses in a short-time scale and question linear relationship between electricity demand and meteorological data. Gained results could help in future energy system planning of the power system with a high share of RES, for the Dubrovnik region, as well as other regions. Correlation analyses of the first set of data provided results of strong correlation for solar radiation, air temperature and electricity demand distribution between selected years, showing low variation of data in between the years. These results indicate the possibility of a good data forecasting using equations of linear regression line in the 10 minute time step, thus contributing to the future energy system planning. Analyses on wind speed data gained weaker correlation results which means they cannot be pronounced with linear regression line. Air temperature is shown to be a reliable factor used in predicting solar radiation and energy demand.

Linear regression and correlation analyses of the second data set, based on the mean monthly values of the selected variables, gained similar values of correlation coefficient as in the previous studies done for other regions. Results indicate very significant relation of I_n *vs.* V_n with $-0.86 < r < -0.81$, meaning that the solar radiation increases as wind speed decreases, which indicates the possibility of combining these two sources in electricity production. 2014 yield representative and significant results for E_{2014} *vs.* I_{2014} with correlation coefficient in value of 0.64 representing solar radiation as a good power source. Correlation of E_n *vs.* V_n with $-0.48 < r < -0.23$ indicated negative correlation and was not significant for each year. This study approved combination of wind and sun in electricity production for the selected region. These RES, along with hydro potential, will be considered as the main energy sources for electricity production in our future work of planning the energy system with a high share of RES.

This work went a step forward from previous studies and provided results of the linear regression and correlation between the selected parameters for the 10 minute time step. Correlation results indicate a decrease in correlation coefficient values with a slight improvement when system time delay was considered. The results were more improved when we analysed the relation between the variables for the summer and winter period separately. Our further work will deal with energy plan models and energy market based on the short-term scale. Since the results of this study indicated weak relations between the variables based on the 10 minute time step, especially for the wind speed data, further work needs to be done in order to enhance the integration of RES in the power system production. Additional flexibility will have to be ensured in order to achieve stability in the power system and to accomplish the targets of the electricity production in Croatia by 2020, mentioned in section 0. Additional flexibility can be achieved through additional storage facilities, optimisation using market electricity prices, implementing ICT-tools, as well as combining electricity, heat and transport sector. Correlation analyses between the data will certainly impact the decrease of flexibility

requirements, but there will still be the need for some flexibility in the system. Correlation and regression results based on the mean monthly data can be helpful in energy system planning to determine the type of sources used in electricity production, their installed capacities and optimal mix. Results based on the 10 minute data showed that the electricity demand, solar radiation and air temperature can be easily forecasted in a short-term scale. Future energy markets will be arranged in a short-term scale, where ICT-tools will help in energy flow regulation which will be controlled by energy prices. Energy prices will depend on the energy production and energy source, and correlation and regression analyses could provide useful results for future smart energy system planning. In our future work we will deal with energy system planning with a high share of RES in electricity production, based on the short-term scale, in order to reduce the pollution and achieve flexibility and stability of energy system.

Acknowledgment

Authors would like to thank Croatian Meteorological and Hydrological Service and Elektrojug Dubrovnik – HEP ODS d.o.o. for providing data for this work.

References

[1] ***, 2009/30/EZ Directive of European Parliament and Council, 2009, http://eur-lex.europa.eu/legal-content/HR/TXT/?uri=celex:32009L0030
[2] ***, Energy Strategy of the Republic of Croatia, Zagreb, June 2009, http://weg.ge/wp-content/uploads/2013/05/Croatia-Energy-Strategy-2009.pdf
[3] Kayal, P., Chanda, C. K., A Multi-objective Approach to Integrate Solar and Wind Energy Sources with Electrical Distribution Network, *Solar Energy*, *112* (2015), Feb., pp. 397-410
[4] Tsekouras, G., Koutsoyiannis, D., Stochastic Analysis and Simulation of Hydrometeorological Processes Associated with Wind and Solar Energy, *Renewable Energy*, *63* (2014), Mar., pp. 624-633
[5] Kayal, P., Chanda, C. K., Optimal Mix of Solar and Wind Distributed Generations Considering Performance Improvement of Electrical Distribution Network, *Renewable Energy*, *75* (2015), Mar., pp. 173-186
[6] Purvins, A., *et al*., Effects of Variable Renewable Power on a Country-scale Electricity System: High Penetration of Hydro Power Plants and Wind Farms in Electricity Generation, *Energy*, *43* (2012), 1, pp. 225-236
[7] Silvente, J., *et al*., A Rolling Horizon Optimization Framework for the Simultaneous Energy Supply and Demand Planning in Microgrids, *Applied Energy*, *155* (2015), Oct., pp. 485-501
[8] Mahesh, A., Sandhu, K. S., Hybrid Wind/photovoltaic Energy System Developments: Critical Review and Findings, *Renewable and Sustainable Energy Review*, *52* (2015), Dec., pp. 1135-1147
[9] Perez-Navarro, A., *et al*., Experimental Verification of Hybrid Renewable Systems as Feasible Energy Sources, *Renewable Energy*, *86* (2016), Feb., pp. 384-391
[10] Mesarić, P., Krajcar, S., Home Demand Side Management Integrated with Electric Vehicles and Renewable Energy Sources, *Energy and Buildings*, *108* (2015), Dec., pp. 1-9
[11] Cai, Z., *et al*., Application of Battery Storage for Compensation of Forecast Errors of Wind Power Generation in 2050, *Energy Procedia*, *73* (2015), June, pp. 208-217
[12] Foley, A. M., *et al*., Current Methods and Advances in Forecasting of Wind Power Generation, *Renewable Energy*, *37* (2012), 1, pp. 1-8
[13] Rasheed, A., *et al*., A Multiscale Wind and Power Forecast System for Wind Farms, *Energy Procedia*, *53* (2014), Sep., pp. 290-299
[14] Wang, X., *et al*., A Review of Wind Power Forecasting Models, *En. Proc.*, *12* (2011), Sep., pp. 770-778
[15] De Giorgi, M. G., *et al*., Assessment of the Benefits of Numerical Weather Predictions in Wind Power Forecasting Based on Statistical Methods, *Energy*, *36* (2011), 7, pp. 3968-3978
[16] Zhao, X., *et al*., Review of Evaluation Criteria and Main Methods of Wind Power Forecasting, *Energy Procedia*, *12* (2011), Dec., pp. 761-769
[17] Alessandrini, S., *et al*., A Novel Application of an Analog Ensemble for Short-term Wind Power Forecasting, *Renewable Energy*, *76* (2015), Apr., pp. 768-781

[18] Hong, Y. Y., *et al*., Hour-ahead Wind Power and Speed Forecasting Using Simultaneous Perturbation Stochastic Approximation (SPSA) Algorithm and Neural Network with Fuzzy Input, *Energy*, *35* (2010), 9, pp. 3870-3876
[19] Alonso-Montesinos, J., Batlles, F. J., Solar Radiation Forecasting in the Short- and Medium-term Under all Sky Conditions, *Energy*, *83* (2015), Apr., pp. 387-393
[20] Gupta, R. A., *et al*., BBO-based Small Autonomous Hybrid Power System Optimization Incorporating Wind Speed and Solar Radiation Forecasting, *Renewable and Sustainable Energy Reviews*, *41* (2015), Jan., pp. 1366-1375
[21] Tascikaraoglu, A., *et al*., An Adaptive Load Dispatching and Forecasting Strategy for a Virtual Power Plant Including Renewable Energy Conversion Units, *Applied Energy*, *119* (2014), Apr., pp. 445-453
[22] He, W., Deep Neural Network Based Load Forecast, *Computer Modelling & New Technologies*, *18* (2014), 3, pp. 258-262
[23] Monforti, F., *et al*., Assessing Complementarity of Wind and Solar Resources for Energy Production in Italy, A Monte Carlo Approach, *Renewable Energy*, *63* (2014), Mar., pp. 576-586
[24] Alham, M. H., *et al*., Optimal Operation of Power System Incorporating Wind Energy with Demand Side Management, *Ain Shams Engineering Journal*, (2015), In press
[25] Guan, H., *et al*., Response of Office Building Electricity Consumption to Urban Weather in Adelaide, South Australia, *Urban Climate*, *10* (2014), 1, pp. 42-55
[26] de Jong, P., *et al*., Solar and Wind Energy Production in Relation to the Electricity Load Curve and Hydroelectricity in the Northeast Region of Brazil, *Renewable and Sustainable Energy Reviews*, *23* (2013), July, pp. 526-535
[27] Huber, M., *et al*., Integration of Wind and Solar Power in Europe, *Energy*, *69* (2014), May, pp. 236-246
[28] Bale, C. S. E., *et al*., Energy and Complexity: New Ways Forward, *Applied Energy*, *138* (2015), Jan., pp. 150-159
[29] Foley, A. M., *et al*., Addressing the Technical and Market Challenges to High Wind Power Integration in Ireland, *Renewable and Sustainable Energy Reviews*, *19* (2013), Mar., pp. 692-703
[30] Jaradat, M., *et al*., The Internet of Energy: Smart Sensor Networks and Big Data Management for Smart Grid, *Procedia Computer Science*, *56* (2015), July, pp. 592-597
[31] Gaviano, A., *et al*., Challenges and Integration of PV and Wind Energy Facilities from a Smart Grid Point of View, *Energy Procedia*, *25* (2012), Sep., pp. 118-125
[32] Mourshed, M., *et al*., Smart Grid Futures: Perspectives on the Integration of Energy and ICT Services, *Energy Procedia*, *75* (2015), Aug., pp. 1132-1137
[33] Lund, H., *et al*., From Electricity Smart Grids to Smart Energy Systems – A Market Operation Based Approach and Understanding, *Energy*, *42* (2012), 1, pp. 96-102
[34] Lund, H., *et al*., Renewable Energy Systems – A Smart Energy Systems Approach to the Choice and Modelling of 100% Renewable Solutions, *Chemical Engineering Transactions*, *39* (2014), May, pp. 1-6
[35] Šare, A., *et al*., The Integration of Renewable Energy Sources and Electric Vehicles Into the Power System of the Dubrovnik Region, *Energy, Sustainability and Society*, *5* (2015), 27, pp. 1-16
[36] ***, Meteorological and Hydrological Service, http://meteo.hr/index_en.php
[37] ***, http://www.kippzonen.com/Product/14/CMP21-Pyranometer#.VvUN59IrLIV
[38] ***, http://www.kippzonen.com/Product/262/CVF4-Ventilation-Unit#.VvUN6NIrLIV
[39] ***, http://www.kippzonen.com/Search.aspx?id=304&query=cm121#gsc.tab=0&gsc.q=cm121&gsc.page=1
[40] ***, Elektrojug Dubrovnik – HEP ODS d.o.o., http://www.hep.hr/ods/en/dp/default.aspx
[41] ***, STATISTICA, http://www.statsoft.com/

Paper submitted: December 9, 2015
Paper revised: April 25, 2016
Paper accepted: April 26, 2016